THE COTANGENT COMPLEX OF A LOCAL COMPLETE INTERSECTION IS PERFECT WITH TOR AMPLITUDE IN DEGREES MINUS ONE AND ZERO

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The two-term presentation of the naive cotangent complex of a quotient of a polynomial algebra. used at proof lines 22864-22869. Provides the representing complex whose amplitude is to be bounded.
- The criterion that a complex with finite free last term has tor amplitude in a range if and only if the previous term is flat. invoked at proof lines 22870-22872. Reduces the amplitude condition to flatness of the conormal module.
- A quasi-regular ideal has finite projective conormal module. invoked at proof lines 22874-22875. Upgrades flatness to finite projectivity, giving perfectness.
- A Koszul regular ideal is quasi-regular. invoked at proof lines 22872-22873. Connects the definition of local complete intersection to the quasi-regularity input.

Source reference. Original definition 8338: sources/stacks-more-algebra/pinned-original-source.tex, source lines 8338–8338. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 16883: sources/stacks-more-algebra/pinned-original-source.tex, source lines 16883–16883. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

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Source reference. Original lemma 8160: sources/stacks-more-algebra/pinned-original-source.tex, source lines 8160–8160. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original section 8092: sources/stacks-more-algebra/pinned-original-source.tex, source lines 8092–8092. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $A \to B$ be a local complete intersection as in Definition 8338. Then $NL_{B/A}$ is a perfect object of $D(B)$ with tor amplitude in $[-1, 0]$.*

Proof. Write $B = A[x_1, \ldots, x_n]/I$. Then $NL_{B/A}$ is represented by the complex

$$I/I^2 \longrightarrow \bigoplus B\mathrm{d}x_i$$

of B-modules with I/I^2 placed in degree -1. Since the term in degree 0 is finite free, this complex has tor-amplitude in $[-1, 0]$ if and only if I/I^2 is a flat B-module, see Lemma 16883. By definition I is a Koszul regular ideal and hence a quasi-regular ideal, see Section 8092. Thus I/I^2 is a finite projective B-module (Lemma 8160) and we conclude both that $NL_{B/A}$ is perfect and that it has tor amplitude in $[-1, 0]$. □